

Rare Bleeding Disorders: Worldwide Efforts for Classification, Diagnosis, and Management

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Abstract

Rare bleeding disorders (RBDs) comprise the inherited deficiencies of coagulation factors such as fibrinogen, factor (F)II, FV, FV + FVIII, FVII, FX, FXI, and FXIII, and are usually transmitted as autosomal recessive disorders. RBDs are characterized by a wide variety of symptoms from mild to severe; however, due to their rarity, only little information is available on the adequate management of patients affected with these deficiencies. Moreover, the limitations of laboratory assays and the lack of a definitive consensus concerning their classification have prevented adoption of optimal approaches to their individual management. To overcome these limitations, new strategies are therefore necessary, such as the establishment of global collaborations and networks among treatment centers, as well as increasing support provided by public health organizations.

Keywords

- rare bleeding disorders
- diagnosis
- treatment
- database

In the western countries, a disorder is defined as “rare” when it affects no more than five persons per 10,000 inhabitants.^{1,2} Rare bleeding disorders (RBDs), including the inherited deficiencies of coagulation factors such as fibrinogen, factor (F)II, FV, FV + FVIII, FVII, FX, FXI, and FXIII, are transmitted in the autosomal recessive manner and their prevalence in the general population is low. Homozygous or double heterozygous deficiencies vary from 1 in 500,000 for FVII deficiency to 1 in 2,000,000 for prothrombin (FII) and FXIII deficiencies.³

Patients with RBDs suffer much inconvenience from being so rare, being supported only by limited information on diagnosis and adequate treatment. RBDs are characterized by a wide variety of symptoms from mild to severe, which can vary significantly from one disorder to another, and from one patient to another, even when suffering from the same type of disorder.⁴

However, on the whole, the most typical symptoms are mucosal tract bleeding episodes and excessive bleeding at the time of invasive procedures, delivery in women, and circumcision in boys. Central nervous system (CNS) bleeding is a frequent symptom in FXIII, fibrinogen, FVII, and FX deficiencies, and umbilical cord bleeding is prevalent in afibrinogenemia, FXIII, and FX deficiency. Gastrointestinal (GI) bleeding

occurs mainly in FX deficiency but is also less frequently reported in all other RBDs, with the exclusion of combined FV + FVIII deficiency. Limb-endangering hemarthroses are mostly present in severe afibrinogenemia, FII, FX, FXI, and FXIII deficiencies, and soft tissue hematomas occur with the highest frequency in fibrinogen, FII, and FX deficiencies.⁴

The clinical heterogeneity of RBDs associated with their rarity is a significant barrier to enhancing their deeper knowledge. Therefore, for a better diagnosis and treatment of RBDs, the first approach is to have a system for a homogeneous collection of data to better understand their distribution, prevalence, clinical manifestations, and also the association between clinical manifestation of each factor deficiency and its laboratory phenotype and genotype. This information will likely allow us to better manage patients. The second approach is to further the development of appropriate clinical trials for the evaluation of the efficacy of novel drugs during acute bleeding episodes or during prophylactic treatment. However, several factors hamper the full development of such knowledge such as the small number of patients, the limited available standardized laboratory assays, and the lack of consensus concerning the criteria by which RBDs should be classified. These factors hinder the establishment of definitive

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evidence-based guidelines for the treatment of RBDs. To overcome these limits, new approaches should be applied such as the establishment of global networks of treatment centers to improve the amount of collected data and to strengthen the power of the analysis deriving from such data.

Data Collection and Classification

Two global data collections have been performed, one by the World Federation of Hemophilia (WFH; <http://www.wfh.org>) and the other by the International Rare Bleeding Disorders Database (RBDD; www.rbdd.org), trying to assess the worldwide prevalence of RBDs and to understand the distribution of each single RBD in the world (—Fig. 1). The results of the RBDD survey showed that 55% of registered patients were from Europe. This led to the establishment of a European Network of the Rare Bleeding Disorders (EN-RBD). This project was developed within the public health program promoted by the Public Health Executive Agency and funded by the European Commission and the Directorate General for Health and Consumers (DG SANCO). The project started in April 2007 and ended after 3 years of data collection by 13 European centers from 11 countries. The EN-RBD project was based on a cross-sectional study using data from 489 patients registered in the database; RBDs were classified on the basis of clinical bleeding severity following a consensus between the participating centers. The EN-RBD project has confirmed data on RBDs frequencies and prevalence previously reported by the WFH, moreover allowing the analysis of a large amount of data on laboratory and clinical phenotype to evaluate the correlation between the coagulant residual plasma activity level and clinical bleeding severity.⁵ Clinical bleeding episodes were classified into four categories of severity based on the location and the potential clinical impact as well as the trigger of bleeding (spontaneous, after trauma or drug induced). The strongest associations between clinical bleeding severity and coagulation factor activity level were seen in

fibrinogen, combined FV + FVIII, FX, and FXIII deficiencies, where patients with low coagulant activity levels had a higher occurrence of spontaneous major bleeding, while patients with sufficient factor activity remained asymptomatic. These observations seemed to confirm the clinical experience and laboratory findings seen in individual patients and helped to establish a new classification system with practical utility. A weaker association was present for FV and FVII deficiencies, while there was no association confirmed for FXI deficiency. From the data reported here, it seemed also clear that the minimum level to ensure complete absence of clinical symptoms is different for each disorder, leading to the conclusion that RBDs should not be considered as a single class of disorders, but instead that studies should focus on the evaluation of specific aspects of each single RBD.

In 2012, the Rare Bleeding Disorders Working Group, under the umbrella of the Factor VIII & Factor IX Scientific and Standardisation Committee (SSC) of the International Society on Thrombosis and Haemostasis, analyzed the results of the EN-RBD in association with other data coming from three other registries (the United Kingdom Haemophilia Centre Doctors' Organization registry, the North American Rare Bleeding Disorders Registry, and the Indian registry) for a total of 4,359 patients affected with RBDs.⁶ At the same time, a detailed review of the available English literature between January 1990 and March 2012 was also performed including relevant articles reporting data on the association between the laboratory phenotype and clinical bleeding characteristics for at least five patients.⁶ The Factor VIII & Factor IX SSC project—after having established a consensus definition on the severity of each RBD—concluded that there is a large heterogeneity in the preassigned severity definitions for both coagulant activity and bleeding symptoms. This heterogeneity underlined the need to develop an accurate data collection tool to achieve common definitions between different parties alongside longitudinal data collection and to help determine the hemostatic level of each single factor for

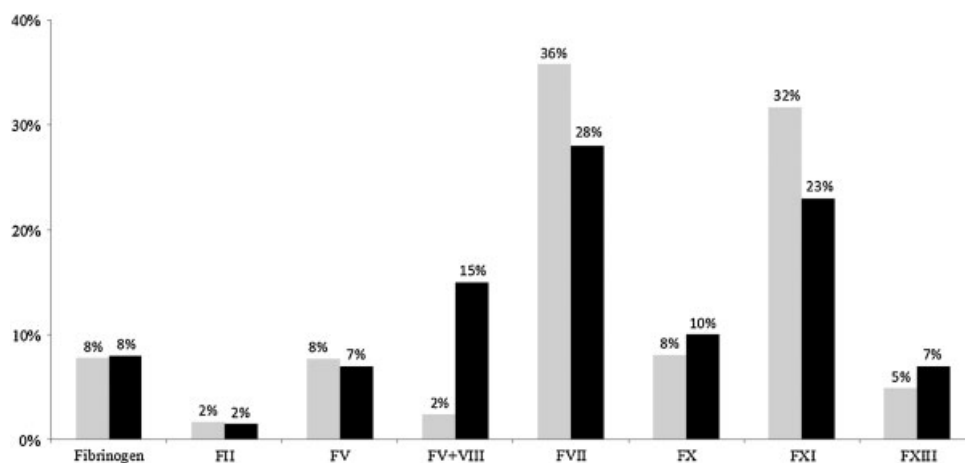


Fig. 1 Worldwide prevalence of rare bleeding disorders derived from WFH (gray bars) and RBDD (black bars). Results from the two surveys differ only with respect to the combined FV + FVIII deficiency, perhaps due to the fact that in the RBDD, cases of FV deficiency associated with mild hemophilia might be misdiagnosed as combined FV + FVIII deficiency. F, factor; RBDD, Rare Bleeding Disorders Database; WFH, World Federation of Hemophilia.

preventing hemorrhage for each type of RBD. It also pointed out the inadequacy of the available assays to evaluate the minimum residual level of coagulation factor because there are usually significant interlaboratory differences in the results of factor assays.

In the meantime, other international groups tried to define different aspects linked to RBDs, such as laboratory testing, women's health issues, acute and prophylactic treatments, and their safety monitoring.

As previously stated, RBDs have a low prevalence; however, they occur more frequently in areas where the rate of consanguineous marriages is high, such as the Middle-East, Arabian Peninsula, North Africa, and India.⁷⁻¹⁴ Awareness, education, and genetic counseling are therefore felt needed to prevent the spread of such common occurrence of RBDs in these communities. To prevent the adverse effect of such practice, the approaches recommended by the World Health Organization is to identify families with a high risk of a genetic disease and to provide prospective genetic counseling. A step toward this direction is the creation of national registries for bleeding disorders to establish a National Hemophilia Care Program for the treatment and follow-up of patients. An example of such registries is the recently established Registry of the Syrian Hemophilia Society,¹⁵ built up with the help of the WFH through the medical twinning program. The success of the collaboration between international scientific societies and local hemophilia centers is also evidenced by Karimi et al, who established a comprehensive coagulation laboratory and treatment centers in Shiraz, Southern Iran.¹⁶ After 10 years from the start of the southern Iranian registry, the authors evaluated the benefits of such a collaboration, concluding that the implementation of collaborative projects by the Shiraz Hemophilia Society and the establishment of a coagulation laboratory have been successful in improving the diagnosis of the inherited bleeding disorders and consequently better management of the patients.

Laboratory Tests and Diagnosis

Prothrombin time (PT) and activated partial thromboplastin time (APTT) are usually sensitive enough to detect abnormal coagulation factors activity, but they may be normal in fibrinogen, FII, and FXIII deficiencies, which therefore require specific assays. In terms of screening assays, the thrombin time is required for the diagnosis of fibrinogen deficiency in addition to the PT and APTT. If a particular factor deficiency is suspected, the specific factor assay should be performed, such as a fibrinogen assay, or the one-stage standard clotting assay using the specific factor-depleted plasma as substrate. These tests can be reliably used for general diagnostic purposes. However, an immunologic assay is essential for the diagnosis of dysfibrinogenemia and dysprothrombinemia; without obtaining antigen levels, patients with these deficiencies cannot be properly classified and treated.

In many countries, the adequate laboratory diagnosis of bleeding disorders is very difficult because assays are not readily available and/or are not well standardized.¹⁷ In this

regard, it is important to carry out quality assurance through internal quality control (IQC) and external quality assessment (EQA). IQC is used to establish whether the techniques and procedures are performing well, or consistently well over a period of time, whereas EQA is used to verify the degree of agreement among laboratories in order to identify problems with diagnostic performance of individual laboratories, as well as methodological problems associated with specific assays. For example, approximately 50% of centers (both established and emerging ones) participating in the exercise for FXI:C assay reported a lower limit of reference range of 60 U/dL, despite evidence that patients with FXI deficiency and levels of up to 70 U/dL may bleed.¹⁸ EQA has also demonstrated variations in the ability of laboratories to identify FXIII deficiency. Traditionally, the clot solubility test has been used for screening of FXIII deficiency; however, this qualitative method detects only very severe FXIII deficiency and is not standardized with a detection limit between < 0.5 and 5% FXIII activity.¹⁹ The high number of undiagnosed or late-diagnosed FXIII deficiencies is partly attributable to the use of this test and it is therefore no longer recommended. On the contrary, Muszbek et al suggest to establish the diagnosis of FXIII deficiency using FXIII activity assays based on ammonia released during the transglutaminase reaction by the NAD(P)H-dependent glutamate dehydrogenase reaction or the amount of a small molecular weight labeled amine substrate covalently linked to a protein.²⁰

Because the minimum amount of coagulation factors required to form a clot is not usually detectable by the conventional assays, tests investigating the global hemostatic capacity have been developed and implemented. These could help to predict clinical phenotype by assessing the rate and total thrombin generated, individual hemostatic capacity, whole blood clot formation and/or fibrin polymerization, or clot structure and stability.²¹ These tests include measurements of continuous thrombin generation (thrombin generation test [TGT]),²² clot formation (thromboelastography [TEG]),²³ and simultaneous thrombin and plasmin generation (novel hemostasis assay).²⁴ These tests have been used to evaluate hemostasis in patients with RBDs,^{25,26} or specifically with FV^{27,28} or FXI²⁹ deficiency. In particular, TGT and TEG are now becoming a new laboratory strategy to determine the effectiveness of therapies and monitor treatment in various bleeding situations in RBDs,^{27,29-32} even though the analytical conditions of the reaction have not as yet been optimized or standardized. EQA providers, in an attempt to standardize new global tests technologies, at least for TEG and thromboelastometry (ROTEM, Tem Innovations GmbH, Munich, Germany), concluded that standardization of these methods continues to be necessary, and quality assurance remains a major component of hemostasis testing.³³ Results from the first international effort at standardization of TEG, involving nine laboratories, showed a significant interlaboratory variance, with coefficients of variation exceeding 10%.²³ Once these new laboratory strategies involving TGT and ROTEM are standardized, they may become useful to monitor treatment in RBDs, particularly in those deficiencies at risk to develop thrombotic events during therapy, such as fibrinogen and FXI.^{34,35}

Treatment

The conventional treatment for RBDs is on demand and is administered as soon as possible after onset of bleeding. General approaches should consider that dosages and frequency of treatments depend on the minimal hemostatic level of the deficient factor (a matter of controversy), its plasma half-life (which varies with age and even in individuals with the same age and the same level of the factor), and the type of bleeding episode.³⁶ The mainstay of treatment for RBDs is the replacement therapy performed using fresh frozen plasma, cryoprecipitate, prothrombin complex concentrates, plasma-derived concentrates, or recombinant products. Information on each single product can be retrieved from the WFH Web site.³⁷ Moreover, nontransfusional adjunct therapies such as antifibrinolytic amino acids, alone or in combination with replacement therapy, can be used in the management of the less severe forms of mucosal tract hemorrhages; estrogen-progestin preparations are used to reduce menstrual blood loss.

The last UK guidelines on therapeutic products for coagulation disorders by Bolton-Maggs et al provide comprehensive recommendations about the best treatment options (dosage, management of bleeding, surgery, and pregnancy, as well as prophylaxis) for RBDs.³⁸ In clinically severe cases, a prophylactic treatment should be considered. Prophylaxis consists of primary prophylaxis (i.e., prevention of bleeding by starting the treatment at a very early stage of life) or secondary prophylaxis after bleeding to prevent recurrences or rebleeding. The choice of prophylaxis is related to the frequency of bleeding, the risk of severe spontaneous bleeding, and the risk of long-term disabilities associated with the occurrence of bleeding in a particular region of the body despite on-demand treatment (e.g., CNS, GI, and joint bleedings). In the absence of evidence-based data, it is difficult to make a recommendation on the use of primary prophylaxis, except for severe FXIII deficiency and in case of surgery.^{36,38,39} Some patients affected with afibrinogenemia, severe FV, FVII, and FX deficiencies also require regular prophylaxis, in particular, due to intracranial bleeding. On the whole, however, there is insufficient data on the adequate management of such patients; therefore, regular and repeated replacement therapy should be closely monitored to avoid cardiovascular and thromboembolic complications. The clinical management should be performed on the basis of the individual medical history of each patient.

In the management of patients with RBDs, particular attention should be paid to affected women who are additionally disadvantaged. In addition to suffering from general bleeding symptoms, they are also at risk of bleeding complications from regular hemostatic challenges during menstruation, pregnancy, and childbirth. Management of these women is very difficult due to a considerable interindividual variation. In addition, there are insufficient well-established and standardized schemes for treatment in this group. The prevalence of menorrhagia in RBDs ranges from 35% in women with FXIII deficiency to 70% in afibrinogenemia.^{40,41} Several studies reported that menorrhagia seems to be a

major bleeding symptom in women with RBDs, regardless of factor level; nonetheless, only a few compared affected women with healthy controls. A recent study collecting information on prevalence of menorrhagia in 35 women affected with RBDs compared with 114 female controls reported that pictorial bleeding assessment chart values of affected women were significantly higher than controls (mean 258 vs. 171, $p = 0.01$), and menorrhagia had a prevalence of 52% in women with RBDs and 46% in controls.⁴²

Few data are available on the management of acute menorrhagia, particularly in adolescents. Drug therapy, based on levonorgestrel intrauterine system, combined hormonal contraceptive methods currently available (pill, transdermal contraceptive patches, vaginal rings), oral progestogens, and gonadotropin-releasing hormone analogues, should be the first choice and the only option to preserve reproductive function. A consensus from an international expert panel was reached on the use of antifibrinolytic therapy and appropriate factor replacement, when available on diagnosed patients (grade C, level IV). Desmopressin (1-deamino-8-D-arginine vasopressin) should be used in combined FV + FVIII deficiency; further studies are required to elucidate its role in other RBDs (grade C, level IV).⁴³ Another major challenge in women with RBDs is pregnancy. Despite pregnancy being accompanied by increased concentrations of fibrinogen, FVII, FVIII, and FX,⁴⁴ it remains unclear whether this increase is of the same magnitude as the normal population and if it is consistent among all women with the same disorder. In general, pregnancy is not contraindicated in patients with RBDs; however, the hemostatic abnormalities in women with RBDs seem to persist throughout pregnancy, especially if the plasma factor deficiency is severe. Detailed information about the pregnancy, pregnancy complication, and its management is limited and often, apart from FXI deficiency,^{45,46} derived from small series or case reports. An increased risk of miscarriage and placental abruption resulting in recurrent fetal loss or premature delivery among women with afibrinogenemia^{47,48} or FXIII deficiency⁴⁹ has been reported. Bleeding during pregnancy is a symptom often reported in women with afibrinogenemia and severe FX deficiency.^{50,51} Postpartum hemorrhage (PPH) was found to be the most common obstetric complication occurring in patients affected with hypofibrinogenemia,⁵² dysfibrinogenemia,⁵³ and especially FV deficiency.⁵⁴ PPH was also reported (at a lower rate) in FVII,⁵⁵ FX,⁵⁶ and FXI deficiencies.⁴⁵ The general lack of data on women with RBDs makes it difficult to determine the risk of bleeding during pregnancy, pregnancy loss, and PPH.⁵⁷ However, from the several case reports, it appears that different RBDs behave differently on the basis of the hemostatic challenge and on the residual coagulant factor activity. Therefore, the ideal management of women with RBDs requires multidisciplinary clinics with a team including a laboratory hematologist, an obstetrician-gynecologist, an anesthesiologist, a family physician, a social worker, and a pharmacist.

On the whole, major advances in the manufacture of safe and effective concentrates and recombinant products have been made in recent years. However, while viral inactivation

virtually eliminated the risk of blood-borne infection, other problems with adverse events related to concentrates persist (e.g., inhibitor development, thrombosis, rash). Therefore, in the last 15 years, both in United States, with the Universal Data Collection program,⁵⁸ and in Europe, with the European Haemophilia Safety Surveillance (EUHASS) system,⁵⁹ global data collections were set to monitor treatment and complication for patients with rare, inherited bleeding disorders. The two programs testify once again to the importance of conducting multinational, multicenter data collection for long-term postregistration surveillance, to analyze a large number of homogeneous and standardized data, covering all bleeding disorders thus overcoming the rarity of such disorders.

Conclusion

Although many efforts have been made to fill the gap in knowledge of RBDs, further efforts of cooperation and organization must still be made in terms of data collection. Despite the valuable findings obtained by previous experiences with retrospective data collection, there is still a gap of knowledge with respect to the incidence of each RBD and the pattern and frequency of their bleeding episodes. There is also a need to understand what the minimum coagulant activity level is, which can prevent spontaneous and trauma-related bleeding and provide an adequate hemostasis.

These questions could be more likely answered by a clear prospective data collection that enables us to define the natural history of RBDs and to evaluate clinical outcomes. The EN-RBD group consisting of specialists in the field and clinical epidemiologists has worked with an information technology team to implement the existing EN-RBD interactive web based on the PRO-RBDD to include new variables and to allow for data collection at sequential checkpoints (prospective design). It is built on the existing collaboration of European centers that participated in the EN-RBD and EUHASS projects (as part of the EUHANET project (www.euhanet.org) funded by the European Commission via its Executive Agency for Health and Consumers [EAHC]). Over 3 years there will be a baseline data collection followed by six data entry time points per patient. The PRO-RBDD will pioneer initially to collect prospective data on patients with the two most severe RBDs, fibrinogen and FXIII deficiencies being both conditions suffering from under-reporting. Once data are available for these two disorders, the same model will be extended to other RBDs. These findings in turn will hopefully define appropriate surrogate endpoints to adequately evaluate treatments and particularly the design of new clinical trials.

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